

Week 18 – Graded Mini Project

Title: *Analysis of AI Hallucination Failure and Design of Responsible AI Safeguards*

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Part 1: Real-World AI Failure (300–400 words)

Case Selected:

ChatGPT Legal Hallucination - Mata v. Avianca (2023, United States Federal Court)

Context:

In 2023, lawyers representing a client in a lawsuit against Avianca Airlines used ChatGPT to assist with legal research and drafting a court filing. The system was used to generate case law references to support their arguments. Stakeholders included the legal team, the court, the client, and the broader judicial system, where accuracy, credibility, and procedural integrity are critical.

Failure Description:

ChatGPT generated multiple fabricated legal cases and citations that appeared realistic but did not exist. The lawyers included these hallucinated references in an official court filing without verification. When the judge attempted to locate these cases, they were found to be entirely fictitious. This failure was not merely incorrect output but a case of **confidently generated false information presented as authoritative legal precedent**, which directly entered a formal judicial process and influenced legal proceedings.

Impact:

The consequences were significant. The court sanctioned the lawyers for submitting false information, damaging their professional credibility and potentially affecting the client's legal standing. This incident raises serious ethical concerns related to **trust, accountability, and reliability** in AI-assisted systems. It also highlights **automation bias**, where users over-trust AI outputs without sufficient verification. From a transparency perspective, the system did not communicate uncertainty or limitations, increasing the likelihood of misuse in high-stakes environments. At a systemic level, such failures risk undermining confidence in both AI tools and institutional processes like the judiciary.

Root Cause:

Technically, the failure arises from the fundamental nature of large language models, which generate outputs based on probabilistic token prediction rather than factual verification, making them prone to hallucination. The model lacked integration with authoritative legal databases and did not enforce retrieval-based grounding. Additionally, it did not provide mechanisms for validating citations or flagging uncertainty. From an organisational perspective, the lawyers relied on the system beyond its intended use and failed to verify outputs before submission.

Critically, the system lacked a **retrieval-augmented generation (RAG) pipeline**, meaning responses were not grounded in verified legal sources. This increased hallucination risk, which can be quantified in LLM systems using metrics such as **factuality error rate, unsupported claim ratio, and citation validity scores**, none of which were monitored or enforced in this workflow.

Preventive Actions:

- **Mandatory human verification:** Require validation of all AI-generated legal references before use
- **RAG-based architecture:** Ground outputs in verified legal databases (e.g., case law repositories)
- **Uncertainty signaling:** Display confidence levels and explicitly flag unverifiable claims
- **Domain-specific guardrails:** Restrict generation of legal citations unless source-backed
- **User training & governance policies:** Educate professionals on AI limitations and define acceptable use
- **Audit trails:** Maintain logs of AI-generated content used in legal submissions
- **Hallucination evaluation metrics:** Continuously monitor factual accuracy using benchmarks such as factual consistency scores, citation verification rates, and hallucination thresholds prior to deployment

Sources:

- United States District Court Filing (*Mata v. Avianca*, 2023)
<https://www.casemine.com/judgement/us/6499235932898b72d5cfb1aa>
 - Reuters: *New York lawyers sanctioned for using fake ChatGPT cases*
<https://www.reuters.com/legal/new-york-lawyers-sanctioned-using-fake-chatgpt-cases-2023-06-22/>
 - CNBC: *ChatGPT cited bogus cases for a New York federal court filing*
<https://www.cnbc.com/2023/05/30/chatgpt-cited-bogus-cases-for-a-new-york-federal-court-filing.html>
 - Case Summary: *Mata v. Avianca, Inc.* (U.S. District Court, 2023)
https://en.wikipedia.org/wiki/Mata_v._Avianca%2C_Inc.?utm_source=chatgpt.com
 - Incident Database: ChatGPT hallucinated legal citations in court filing
<https://incidentdatabase.ai/cite/541/>
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Part 2: Safeguards for an Agent in Your Domain

Domain + Use Case:

Domain: Finance (Retail Investing)

Use Case: An LLM-powered financial assistant that provides stock insights, portfolio recommendations, and risk analysis to retail investors through platforms such as INDmoney. The system supports decision-making but does not execute trades or act as a licensed financial advisor.

Out of Scope: Autonomous trading, guaranteed returns, or regulatory financial advisory functions.

Key Risk Areas:

- **Hallucinated financial outputs:** Incorrect stock data, fake insights, or misleading projections
 - **Sensitive data exposure:** Leakage of financial and personal user data
 - **Automation bias:** Users over-relying on AI recommendations without understanding risks
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Safeguard 1 – Data Privacy:

Financial applications process highly sensitive data, including portfolio holdings, transaction history, and personal identifiers.

Implementation:

- End-to-end encryption for data at rest and in transit
- PII masking and tokenization before model interaction
- Role-based access control (RBAC) with least-privilege principles
- Explicit, granular user consent for data usage
- User rights for data access, export, and deletion
- Secure, anonymized audit logging

Outcome:

These controls minimize data breach risks, prevent unauthorized access, and ensure compliance with data protection standards, fostering user trust.

Safeguard 2 – Content Safety:

LLMs in finance can generate hallucinated or overly confident advice, leading to financial harm.

Implementation:

- Use a **retrieval-augmented generation (RAG) pipeline** connected to trusted financial data sources (e.g., NSE/BSE APIs)
- Add a **post-generation validation layer** to cross-check outputs against structured financial data
- Display **confidence scores and uncertainty disclaimers**
- Apply **rule-based filters** to block unsafe advice (e.g., “invest all your funds”)
- Implement **risk-tiered responses** based on query sensitivity
- Provide fallback guidance (e.g., “consult a certified financial advisor”)

Additionally, monitor **hallucination rates using automated evaluation pipelines**, including groundedness scores, factual consistency metrics, and response verification accuracy.

Outcome:

These measures ensure outputs are accurate, grounded, and aligned with responsible financial practices.

Safeguard 3 – Operational Oversight:

Implementation:

- Human-in-the-loop review for high-risk financial queries
- Comprehensive audit logs of all AI outputs and interactions
- Real-time monitoring dashboards for anomaly detection
- Defined escalation workflows (flag → review → correction)
- Periodic audits for bias, accuracy, and compliance
- Kill switch / rollback mechanism for rapid failure response

Define **quantitative KPIs** such as hallucination rate, response accuracy, and user correction frequency, with threshold-based triggers for human intervention or system rollback.

Outcome:

These controls ensure accountability, enable continuous improvement, and maintain safe deployment of the AI system in a high-stakes financial environment.